

Socha Lab Project Report

Assembly and Operational Standardization of the Leica RM2125RTS Rotary Microtome.

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Timeline: Aug-Dec, 2021

Objective: Assembly and operation of Leica RM2125RTS rotary microtome; & establishing chemical fixation and embedding protocols.

Preamble:

- Microtome – A specialized precision cutting instrument, to accurately and repeatedly slice sections from a block of embedded tissue, that are later put under light or electron microscopes.
- Leica RM2125RTS – The microtome in Socha Lab is rotational in type that must be operated manually. Section thickness range of it is 5–60.



Figure: Fully assembled rotary microtome (Leica RM2125RTS) at Socha Lab

Accomplished during project:

1. Microtome Assembly:
 - ✓ The instrument and all its parts are currently assembled at Socha Lab (See Figure).

2. Training on Operation:
 - ✓ As Leica's official trainings are available only after January, online video resources have been used for the self-training to operate the microtome properly.
3. Decisions on Chemical Fixation & Embedding Protocol:
 - ✓ Workshopped online with – Dr. Yael Politi – TU Dresden, Germany
 - ✓ Collected resources from:
 - i. Dr. Guenther Pass – University Of Vienna, Austria
 - ii. Dr. Yael Politi – TU Dresden, Germany
 - iii. Dr. Socha & Dr. Khaled Adjerid – Virginia Tech
 - ✓ Based on our decision we ordered:
 - i. Chemicals – Bouin's fixative, Spurr resin, and Ethanol.
 - ii. Equipment – Embedding cassettes and base molds.
4. How the setup and protocols would help:
 - ✓ As part of the primary NSF project on insect respiration, we aim to set the station at Socha Lab to be self-sufficient to conduct all the steps starting from fixation, to slicing for microscopy.